A Review of Noise Production and Mitigation in UAVs

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CONTENTS

The use of Unmanned Aerial Vehicles (UAVs) has increased in the last years in applications such as video entertainment production, search and rescue, security, etc. There has also been a growing interest in analyzing the noise the rotors of such vehicles produce. In this chapter, a review of the literature surrounding the topic of noise produced by UAVs is presented with the following objectives: 1) to describe the recorded effects it has on society; 2) to summarize the results of past audio analysis of noise both in terms of the recorded amplitude of popular UAV models, as well as in terms of its spectral composition and modeling; 3) to review recent noise mitigation techniques. We conclude with a set of insights extracted from the review itself, as well as potential research avenues that may interest the reader.

12.1 Introduction

The growth of the drone market for leisure and civilian applications has raised concerns regarding the public acceptance of drone technology. These concerns include safety, health, animal well-being, and noise pollution issues. The lack of specific standards for drone noise underscores the need to establish means to measure its effects and develop guidelines for deployment in urban settings.

Public acceptance of drones is affected by factors such as annoyance caused by sound levels, societal safety, and environmental impacts. The continuous presence of noisy drones flying over homes and workplaces may lead to decreased public acceptance due to safety and health concerns, similar to living or working near airports.

In this context, this chapter aims to explore the aforementioned issues, touching on three main subjects: 1) the impact of UAV noise on society, 2) noise modeling, and 3) noise mitigation techniques.

Each section aims to provide the reader with a concise description of recent works delving into each subject. From this review, we argue that the topic of noise produced by UAVs or drones (as they are commonly known) is an area with potential research and development opportunities seeking to achieve the concept of Urban Air Mobility (UAM), for which drones are the fundamental element. In this context, drones are envisaged as commercial products for different civilian applications, from catastrophe response, research, and civil protection, to parcel delivery and air taxis.

However, such applications are questioned by the public around the world due to different concerns, namely, criminal activities, misuse, and privacy breaching. Even applications such as parcel delivery and air taxi, with clear benefits to individuals, are perceived with skepticism. In fact, in the case of air taxis, a human pilot is demanded by the users before thinking of making use of such technology.

The above has to be faced on different fronts and one of the most obvious is that of public acceptance, which will increase if concerns around commercial applications are mitigated. This implies the development of guidelines, regulations, liability and insurance policies, and mitigation mechanisms to guarantee the safe use of drones in civilian spaces.

From our review, we noted that public acceptance tends to see as positive those applications that are non-profitable, such as emergency response, research, and civil protection. More advanced commercial applications are also well received, as long as flight operations are not performed near to their home or workplace. This is known as the "not in my backyard" response, which is strongly related to safety, liability, and privacy concerns.

We will also discuss that, even when drone noise does not seem to be a priority, it becomes relevant if the UAM concept becomes a reality with the deployment of hundred or thousand of drones whose collective noise may become worse than urban traffic or aircraft noise in airports. In this respect, drone noise may become a health risk to the public, and therefore, it is essential to develop studies to understand the impact of noise on the public. However, adequate metrics and experimental frameworks must be developed for the latter to enable a thorough evaluation.

Finally, we also discuss noise mitigation techniques, which could enable the development of friendly drones that could be deployed seamlessly in urban applications without calling the attention of civilians. For instance, drones without noise could be used in sports and social events and even for studies of wildlife, since it has also been demonstrated that animals note the presence of drones and may become stressed by them.

We conclude the chapter with a discussion and final remarks, reflecting on the fact that the study of drone noise will contribute to the acceptance of drone technology by the public.

12.2 UAV Noise and its Impact on Society

The growth of the drone market for leisure and civilian applications, namely, aerial video recording and photography, precision agriculture, and outdoor monitoring, among many others, has raised concerns regarding population acceptance of this technology. Several issues arise when Unmanned Aerial Vehicles, known informally as drones, from safety concerns to health and pollution issues from the noise produced during flight. Motivated by these issues, in this section, we will briefly revise relevant works on drone noise and its effect on public acceptance, including the impact of noise on wildlife, including some proposals to evaluate and measure more effectively the effect of noise on the public.

First, it is important to mention that one of the main indicators, mentioned in several of the works in the literature, to attempt to measure a drone acoustic is that of Sound Pressure Level (SPL), see Section 12.3, equation 12.1.

Sound pressure level (SPL) measures the intensity of sound waves relative to the minimum threshold of human hearing, using decibels (dB) as a unit. SPL is a logarithmic scale, this is, every increase of 10 dB represents a ten-fold increase in sound pressure.

Sound pressure levels are often used to describe the loudness of sounds in various environments, such as concerts, workplaces, and residential areas. Exposure to high sound pressure levels can have detrimental effects on human hearing, so it is important to monitor and control noise levels in various settings to prevent hearing damage. Some SPL and their corresponding sources are indicated in Table 12.2 [38]:

Drones are a relatively new source of noise in the environment. In 2019, 1.32 million small UAVs (weighing less than 25 kg) were registered for leisure,

TABLE 12.1
Noise Levels from Different Sources taken from [38]

Sound Source (Noise)	SPL dB	Level
Aircraft at take off	120	Extremely Loud
Car horn	110	Extremely Loud
Subway	100	Very Loud
Truck, motorcycle	90	Very Loud
Busy crossroads	80	Very Loud
Noise level near a motorway	70	Loud
Busy street through open windows	60	Moderate
Light traffic	50	Faint
Quiet room	[40 30 20]	Faint
Desert	10	Faint
Hearing threshold	0	Faint

while 385 thousand were registered for professional applications in the USA. It is forecasted that by 2024, registered drones will increase by 12% (to 1.48 million) for leisure and 115% (to 0.83 million) for professional applications. The ANSI Standardization Roadmap of 2020 [21] highlighted that no specific standards for drone noise currently exist.

This underscores the significance of establishing means to measure the effects of drones and developing guidelines for their deployment in urban settings. Addressing this issue is the primary objective of the study presented in [39]. The article highlights that only a handful of research teams have conducted empirical testing on the short-term impacts of drones on humans, despite the possibility of their acoustic or visual features adversely affecting people's well-being. Also, it is stated that discomfort towards drone noise strongly depends on the SPL as it happens with other transportation noise sources. This work highlights several key points: 1) on average, drones are more annoying than road vehicles in terms of sound pressure level (SPL); 2) hovering multi-rotor drones cause more annoyance than starting jet aircraft; and 3) soundscapes can increase annoyance even when road traffic noise is high. The SPL of a drone depends on its type, size, weight, number of propellers, and distance from the observer, ranging from 60 to over 100 dB. Therefore, the authors in [39] call for more comprehensive evaluations across a wider range of urban scenarios to better understand the noise annoyance caused by drones.

Apart from annoyance caused by sound levels, there are other concerns that affect the public acceptance of drones. These include societal safety and environmental impacts, which have been studied in [10] through various surveys in different countries, and keeping in mind the new concept of UAM, which considers the use of drones for commercial purposes heavily oriented to civilian applications. While the surveys do not indicate any clear trend in the level of acceptance of drones and urban air mobility from 2015 to 2021, the latest survey shows an 83% acceptance rate, the highest yet, as opposed to less than 63% in previous years. However, this rate is not conclusive for commercial use. The surveys reveal that most Americans find the concept of drone delivery attractive, but a significant portion are undecided, and different groups have varying levels of interest in drone delivery. The most pressing issues include drone malfunctions, misuse, privacy concerns, potential damage, and nuisance. The results of surveys also indicate that the costs of drone services and the amount of time saved have a significant impact on respondents' attitudes towards drones. Drone operations related to health and welfare generally receive higher levels of acceptance than those related to leisure or business.

Public opinions on drones vary over time and across countries. However, there seems to be a growing trend of acceptance for urban air mobility, with around half to three-quarters of the public supporting this idea. It is worth noting that people tend to react positively to business models involving drones, provided that safety is ensured and the drones are deployed away from residential or workplace areas, which is often referred to as the "not in my backyard" response. From [10], we can summarize a list of societal concerns:

- **Environment:** noise impact; emissions impact; impacts on animals and flora; recycling; impact of climate change; visual pollution; loss of privacy/intrusion; nuisance.
- **Safety:** safety concerns; security concerns; cybersecurity.
- **Fairness:** lack of transparency; cost of services; competency; liability.
- **Economy:** jobs; economic viability; demand.

Public opinions on drones vary across time and countries, but there is a growing trend of acceptance for urban air mobility, with between half and three-quarters of the public supporting this idea. However, people generally react positively to business models involving drones only if safety is ensured and the drones are deployed away from residential or workplace areas, a response commonly known as "not in my backyard". According to [10], societal concerns related to drones include noise impact, impacts on animals and flora, visual pollution, and loss of privacy or intrusion. The same work provides a list of mitigations to address these concerns, such as avoiding or limiting hovering drone flights, establishing no-fly zones and times, registering electronic equipment, obtaining flight permits and insurance, and implementing similar mitigation strategies used for other urban and aerial transports.

Similarly, the works in [12, 17] evaluated public acceptance of civilian drones for daily-life applications in a study conducted in Germany in 2018. Notably, the "not in my backyard" response was also observed among German participants in the study. The initial insights from the study revealed that applications such as package delivery or aerial taxis are compelling as long as they do not operate above residential areas. However, there were strong concerns about the increase in drone operations.

The study results showed the following general attitudes towards civil drones in Germany: 49% positive, 43% negative, and 8% undecided. It is noteworthy that the positive attitude is only slightly in favor, which could easily change after negative news involving drones. The respondents in the study expressed concerns about civil applications of drones in the following proportions, as shown in Table 12.2. Note that the most pressing concern is the potential for misuse for criminal activities, followed by privacy breaches, liability, safety and damages, and lastly, noise. The low level of concern regarding noise may reflect a lack of awareness about the noise annoyance produced by drones, which is not surprising given that drones are currently not as ubiquitous as other technologies such as mobile phones, GPS, or the internet.

TABLE 12.2
Concerns about civil applications of drones

Concern	**Percentage**
Noise	53%
Animal welfare	68%
Damages and injuries	72%
Transport safety	73%
Liability and insurance	75%
Violation of privacy	86%
Crime and misuse	91%

Table 12.3 provides a breakdown of the participants' attitudes towards various drone applications. In summary, the most accepted applications are those related to catastrophe response, research work, and civil protection, which are activities that may be seen as non-profitable in commercial terms. However, as the applications move towards a business model, the acceptance level decreases. Surprisingly, parcel delivery is one of the least popular applications. This may be related to the "not in my backyard" response, where an individual may recognize the advantages of aerial parcel delivery but also perceive it as a vulnerable activity that could raise concerns similar to those listed in Table 12.2.

It is important to note that even when the drone applications are for personal purposes, the acceptance trends remain the same. The acceptance percentages for five specific applications that the participants would use personally are shown in Table 12.4. The applications related to non-profit or social benefit receive a higher approval, but still not exceeding 50% as shown in Table 12.3, and approval rates decrease as the applications move towards a business model. Air taxi applications receive the largest rejection, which is understandable as people prioritize safety when traveling. The "not in my backyard" response reflects an attitude towards maintaining safety and privacy in their personal space.

In summary, the work described in [12, 17] found that 4 out of 10 respondents had a negative attitude towards civil drones, while 5 out of 10 had a positive attitude, with the rest being undecided. Attitudes were influenced by

TABLE 12.3
Attitudes towards different drone applications (DNK: Do Not Know; DF: Disagree Fully; DS: Disagree Slightly; AS: Agree Slightly, AF: Agree Fully.). Numbers are in percentage (%).

Applications	DNK	DF	DS	AS	AF
Catastrophe response	2	3	3	27	65
Science and Research	0	3	6	37	54
Life-saving and civil protection	0	5	8	27	60
Traffic, surveillance, infrastructure monitoring	3	7	11	36	43
Medicine, transport of samples	2	8	15	26	49
Farming	3	12	18	32	35
Pictures and videos (news)	2	18	22	39	19
Leisure time activities	0	23	30	34	13
Parcel Delivery	2	27	32	25	14
Pictures and videos (advertisement)	1	44	29	17	9

TABLE 12.4
Attitudes towards different drone applications for own purposes (DNK: Do Not Know; DF: Disagree Fully; DS: Disagree Slightly; AS: Agree Slightly, AF: Agree Fully). Numbers are in percentage (%).

Applications	DNK	DF	DS	AS	AF
Protection by police, fire brigades	1	7	9	34	49
First aid	2	11	10	31	46
Parcel delivery	0	44	27	17	12
Leisure time activity	0	47	28	15	10
Air taxi	1	55	30	9	5

gender, age, and level of information. Noise concerns, especially among women, were confirmed as important factors for acceptance. Advanced air mobility for passenger transport had little acceptance, with 85% of respondents indicating they would not use air taxis, but if used, a pilot should be present, which is very relevant for autonomous drone applications given that such an opinion could affect the funding of this type of research. Establishing regulations for noise management is a fundamental challenge for sustainable UAM. Further research should continue to propose relevant regulation.

Although previous research indicated that noise was one of the least concerning issues related to drones, it is important to note that the deployment of noisy drones in urban areas may have negative implications. While a single drone may not be an issue, the continuous presence of many drones flying over homes and workplaces may lead to decreased public acceptance due to safety and health concerns, similar to the impact of living or working near airports. Therefore, it is important to consider the noise impact of drones when designing and deploying them in urban areas.

The World Health Organization (WHO) has found significant evidence where aircraft noise is linked to cardiovascular disease, sleep disturbance, annoyance, and cognitive impairment. They strongly recommend avoiding exposure to levels above 45 dB, as this level has been associated with adverse health effects. It is also recommended to avoid night-time exposure to more than 40 dB, as it has been linked to adverse effects on sleep [31].

The concern over drone noise has prompted research groups to develop studies that call for the creation of evaluation frameworks and metrics. These frameworks aim to understand the effects of drone noise on humans under various scenarios such as urban, rural, and in different soundscapes, and under different drone deployment numbers. Several studies have been conducted in this area [43, 35, 44, 45]. Some researchers have even developed studies in virtual environments to simulate the deployment of thousands of drones, which could provide insight into potential noise effects in large areas [6].

Lastly, we highlight the research conducted in [26], which aimed to investigate the impact of drone noise on the behavior of 18 different species of megafauna. The findings revealed that the different species have varying levels of sensitivity to noise at different frequencies generated by drones. This indicates that drone noise can potentially cause behavioral changes in animals, highlighting the need to reduce drone impact on target species and to inform future drone use guidelines. This is a significant aspect that matters to the public as urban and rural areas are home to various animal species, including pets and strays, whose well-being can also have an impact on the public. Additionally, animal welfare is widely promoted, making it essential to consider their protection and care in drone operations.

12.3 Noise Modeling

As presented in Section 12.2, there is a clear impact of the noise emanating from UAVs towards humans from both the psychological and physiological standpoint, as well as other animal species. It is then of interest to get into how such noise is produced, so as to be able to mitigate it (the topic of Section 12.4).

Conventional UAV noise can be separated into several components [19]:

- The vibration of the UAV inner components.
- The flow of air through the UAV, pushed by the propellers.
- The air turbulence created by the propellers.

The "vibration" component is usually considered considerably smaller compared to the other two components, to the point that is rarely even mentioned. In fact, of the cited references, only [46] mentions it, and concludes that it can just be ignored.

As for the other two components, it appears that the "flow of air" component tends to not be as "annoying" than the "air turbulence" component [39]. As such, the air turbulence caused by the UAV's propellers, at least from what we can gather from the literature, is the main culprit of its noise and, thus, the one with the most effort behind its modeling and mitigation. To this effect, in this work, we will refer to this component as "UAV noise" as a whole, with the understanding that the other two are present but not as imminent to be mitigated.

Several studies [36, 39, 22, 8, 35, 49, 16, 24, 6, 44, 11] have come to the conclusion that UAV noise has the following characteristics:

- **Tonal**. A motor by itself, with its accompanying blades, produces noise that bears one tone: its fundamental frequency (ω_F). Although other tones (or frequencies) are present, the vast majority of which are harmonics located at frequencies $\omega_{h_i} = i * \omega_F$ (where $i > 1$ is an integer). The other frequencies may be attributed to the other aforementioned components, and not as prominent.

- **Broadband.** As it can be trivially deduced, ω_F varies depending on the speed the motor is rotating. However, even though ω_F can range from low- to mid-frequency tones, its harmonics occupy a wide range of high-frequencies (up to a 2 orders of magnitude greater than ω_F), all of which bearing a considerable amount of power, decreasing to a level barely below the power at ω_F.

These characteristics in conjunction, result in a loud and "whiny" sound that, as mentioned before, is considered very "annoying" to humans, even when it's not as loud as other types of urban noises [39].

Furthermore, the power and tonality of the UAV noise appears to be affected by several variables. One of which, as previously mentioned, is the *speed of the motor* (measured in rotations per minute or RPM). It has been successfully used [49] to create a noise emission model based on multiple regression approach, where the power of the whole UAV noise is divided by the contribution of each motor based on its RPM at each moment in time. Another variable is the *number of propellers*, which minimizes the power of the noise the more they are [14]. However, the authors of [9] concluded that even though decreasing the number of blades does in fact increase the power of the noise (since the motor requires to turn faster to create the same amount of thrust), it shifts that power to lower frequencies, making it less "annoying". Yet another variable is the *shape of propellers*, the design of which have been inspired by owl feathers [29, 48], optimized outright with different parameter constraints [14], specifically to modify the tonality and power the UAV noise, and even ones that shapes can be modified on-the-fly [8].

It is important to note that while UAV noise is cumbersome, because of the impact of its motor speed, number and shape of its propellers to its tonality, the combination of all these variables can also be considered unique to an specific UAV model. Meaning, the UAV noise can actually be used as a type of an acoustic fingerprint [34] just by extracting its frequency cepstral features. In fact, as it will be discussed later, even its own tonality can be used as part of a process to mitigate it [28].

There have several works that attempt to model how UAV noise is created:

- [49]: as mentioned earlier, it is based on integrating the noise from several motors, each modeled based on its RPM throughout time.
- [11]: it integrates the frequency-domain version of the Ffowcs-William and Hawkings equations (for drone noise prediction), with an aerodynamic model based on Blade Element Momentum Theory.
- [23], based on Lattice Boltzmann method (LBM) with a simple mathematical ellipsoid model for steady loading noise; unsteady loading noises were not able to be accurately predicted. A similar model was also proposed in [32].

Although all these models were moderately successful in modeling the creation of UAV noise, it is also important to model how this noise propagates through the air. It is well established [19] that the magnitude of the sound pressure level (SPL) can be defined as $L = 10 log \frac{p}{p_r}$ in decibels (dB), where p is the sound pressure magnitude of the acoustic signal, and p_r is a reference pressure of $2 \times 10^{-5} \frac{N}{m^2}$. The SPL of a propagated signal $L(r)$ at a distance r from the source of such signal, can be modeled as:

$$L(r) = L_0 e^{-\frac{\alpha(\omega_F) r}{2}} \tag{12.1}$$

where L_0 is the SPL of the signal at its source. Thus, the farther we are from the source, the less SPL the signal has. Additionally, the air in which the signal propagates acts as a type of low pass filter, since higher frequencies tend to be absorbed by the vibration of its nitrogen and oxygen molecules. This is represented as the attenuation factor ($\alpha(\omega_F)$) that imposes further SPL diminishment as the fundamental frequency of the signal increases.

Of course, there are other elements at play, like reflections from objects in the UAVs surroundings (such as buildings or even the floor itself) or other noise sources, which may increase or decrease the captured SPL $L(r)$. All of this is incorporated in ENVironmental Acoustic Ray-tracing Code (EnvARC) that was used in [6] to simulate the noise of UAVs flying through a simulated urban area, and worked well.

In addition to the distance to the source, the measured SPL is also affected by one's vertical direction in relation to the UAV (aka. elevation angle). In [39], the authors integrate the findings of two other works, and concluded that: 1) the noise SPL increased towards the top and bottom of the UAV (meaning, as the absolute value of the elevation angle increased); and 2) the noise-SPL-to-elevation ratio increased as the noise ω_F increased.

The UAV's pitch angle is also to be considered, since when it is at 0^o, both the pitch and the elevation angle share the same frame of reference. However, when the UAV is moving, caused by a difference in speed between its motors, the UAV's pitch changes and, in turn, the elevation angle as well. To this effect, in [33], the authors found a similar positive relation between noise SPL and pitch.

Interestingly, the authors of [4] found that, from the source, the noise emission is greater on elevation angles closer to the middle. This may seem as contrary to what was found in [39, 33], however, it is important to remember that the SPL noise emitted from the source may not be indicative to the SPL that is measured from a distance. For example, a lower elevation angle is closer to the motors and, thus, it is expected to be louder. On the other hand, measuring the SPL from a distance towards the bottom of the UAV should be louder than from the sides since air is being pushed towards that direction, contributing to a higher SPL.

Finally, it is also important to mention that SPL is not the only manner in which one can measure the impact of UAV noise. As it was mentioned earlier, UAV noise is not actually louder than other types urban noise (such as traffic or airplanes), but it is considered more "annoying" [39]. To this effect, the authors of [45] found that measuring the noise in terms of Tonality, as well as a combination of Loudness and Sharpness, presented a closer match to a rating of a group of human listeners. This is consistent to the aforementioned tonal nature of UAV noise as being a prime characteristic of its unseemliness.

12.4 Noise Mitigation Techniques

As it can be deduced, mitigating UAV noise is of great interest, given that most of the backlash it causes to all the scenarios they could be employed, limiting its applicability. Having described the different characteristics of UAV noise and the efforts behind modeling it, we will now focus in presenting an overview of current approaches to mitigating it.

From what we could gather from the literature, there seem to be four types of mitigation techniques:

1. Trajectory optimization
2. UAV and motor blade design
3. Automatic noise canceling

Their advantages and disadvantages will be discussed specifically in this section, and generally in Section 12.5.

It is important to note that the World Health Organization, guided by the standard ISO 1996-1:2016 (section 3.6.4 of ISO 2016), has established a "day-evening-night-weighted sound pressure level" (L_{DEN}) [13]. This can be used as a type of threshold [20], or an objective to reach for all these techniques.

12.4.1 Trajectory Optimization

Since the SPL of a noise source diminishes when the distance to such source increases, flight paths can be designed such its impact is reduced. In this regard, two mitigation processes can be employed [19]:

1. *Arrival and Departure*. With the types of air-crafts that are able to fly sufficiently high that its noise its negligible, it is really on the arrival and departure phases of its flight when noise mitigation should be considered. To this effect, airport placement, and the design of its respective arrival and departure paths, can consider the terrain/elevation of its surroundings, as well as its demographic makeup to mitigate noise.
2. *Whole Trajectory*. As for other types of air-crafts, such as UAVs, where its flight height is not enough to mitigate its noise, the whole trajectory is to be optimized as to mitigate noise. To do this, it is required to have *a-priori* knowledge of land use by a population that is aimed to not be affected, as exemplified in Figure 12.1.

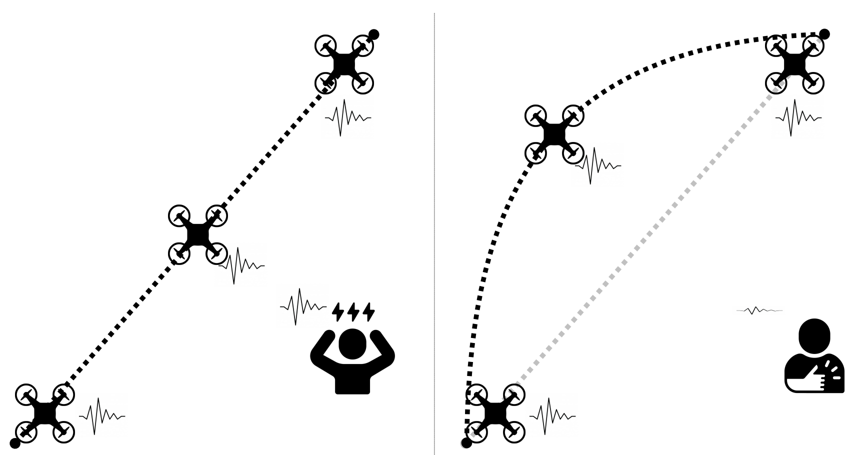

FIGURE 12.1
Example of trajectory optimization. Left: drone is too close to affected individual. Right: trajectory is changed so as to reduce noise impact.

As it can be deduced, the vast majority of flight-path-design-based noise mitigation found in the literature is focused on the second process. A good example of this is the work in [20], which uses the iNoise software [1] to simulate noise to calculate noise (following the standard ISO 9613), and using a rough estimate of land use from an artificially chosen town, provided three flight paths. However, all three flight paths exceeded WHO's L_{DEN} value, even with a small quad-copter.

The work presented in [2] reduces the flying space to an environment, draws a grid over it, and establishes obstacles within in, as well as "quiet zones", areas that should be noise-free. Then, the A* algorithm is used to create different paths, each with a different level of 'quietness'. Another similar work is in [50], where the A* algorithm is modified (into what the authors refer to FairNoiseA*), where the cost of node transitioning is weighted under the amount of noise that will be generated by such transition.

The work in [41] uses agent-based modeling so that UAVs avoid paths that are currently heavily used by other UAVs and, thus, can be assumed that are noisy. Interestingly, this work uses a grid-based trajectory optimization as well, however, because of its complex-system nature, a comparison to A* is difficult.

As for the work in [6], it has an important similarity to the work [2], but uses ENVironmental Acoustic Ray-tracing Code (EnvARC) to simulate noise in a seemingly more sophisticated fashion. Some paths were proposed in a given flight space, however, no trajectory optimization was presented.

It is important to mention that an important limitation of this type of noise mitigation techniques is that the location of quiet zones need to be known *a-priori*. This can be difficult to obtain in some scenarios, such as human activity changing throughout the day. An example of this can be an industry park or a university, both of which are usually not occupied during late evening but are heavily occupied in the mornings and afternoons. Additionally, as established in [41], UAVs increase noise activity once present in an acoustic environments but decrease once they leave it. Thus, trajectory optimization requires up-to-date knowledge of the environment that can be highly dynamic and may not be available.

12.4.2 UAV and Motor Blade Design

By far, this is the type of technique that is the most used for UAV noise mitigation, and can be divided in two parts:

1. Modification of the whole UAV
2. Blade design

The first parts involves the modification of the structure of the UAV to reduce its noise emission. The work in [27] present different manners of sound-proofing the UAV, specifically by the use of rubber, foam pads and aluminium foil. It was found that foam pads helped reduce the noise by a considerable margin (around 15 dB). The work in [40] proposes a UAV with the shape of flying saucer. By employing the Coanda effect [15], which states that a fluid tends to "stick" to a convex surface, the number of propellers are reduced to just one, while still creating enough thrust to lift the UAV.

However, the vast majority of this type of techniques involve the design and number of blades in UAV propellers, as exemplified in Figure 12.2.

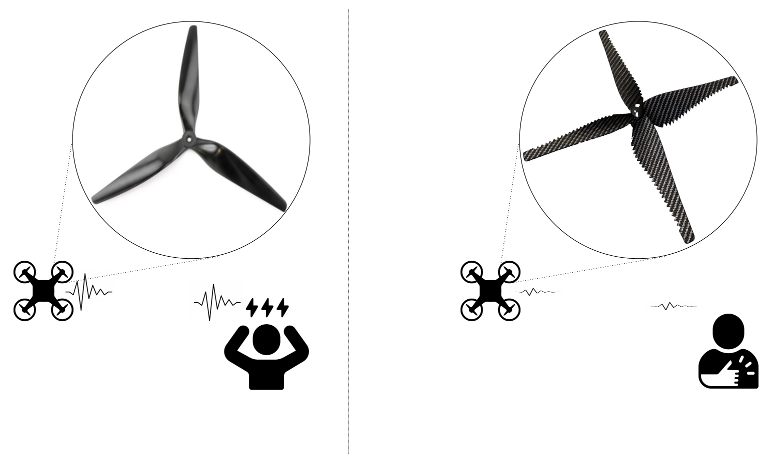

FIGURE 12.2
Example of blade design. Left: drone is using 'noisy' blade design. Right: drone is using 'silent' feather-like blade design.

There is a considerable amount of objectives to optimize for when designing a propeller blade, such as power constraints, thrust production and, of course, noise mitigation. In the work of [14], three different optimization methods (a genetic algorithm, a simplex scheme, and steepest descent) were used to design a blade that maximizes flight and mitigates noise. The authors found that these two objectives were, in a sense, contradictory: blades that maximize flight tended tended to be the loudest. However, they also found that as the number of blades increased, the motor speed requirements to maintain lift decreased, which, in turn, decreased noise.

Biologically-inspired blade designs are quite common, such as using the serrated shape of an owl feather, which the authors of [8, 30, 47, 48] found to be effective in noise mitigation. Other biologically inspired were used, such as the shape of cicada wings and maple seeds, all of which provided similar thrust to a baseline propeller with significantly less noise.

Additionally, the authors in [8] also found that porous material is able to reduce turbulence and, thus, noise emissions.

In [42], several types of propellers were evaluated and found that "loop-type" blade provided the most noise reduction as well as the best performance in psychoactive annoyance level tests.

The authors of [29] proposed the addition of attachments they call 'Gurney flaps' that add a type of serration in the bottom part of the blade, which mitigated the emitted noise at around 1 dB.

In addition to the shape of the blades, there is also the matter of the number of blades to use at each propeller. The authors in [9] found that although increasing the number of blades reduces the level of the noise (which concurs to the findings of [14]), reducing the number of blades 'shifts' the noise to frequencies where human hearing is not as sensitive and, thus, diminishing the 'annoyance' factor. Additionally, the authors of [18] also found that using counter-rotate motors (with the same number of blades) are able to reduce noise while maintaining thrust. In addition, the authors propose running both motors at different speeds to 'spread out' the noise into different frequency bins, reducing the annoyance factor and overall noise level.

It is important to note that the amount of noise mitigated using these blade-design-based techniques is still quite low. Of the works described in this section, the noise reduction reported was between 2 and 5 dB, with a minimum of 1 dB in [29] and up to a maximum of around 10 dB in [47]. For the latter, it should be noted that this reduction was also due to the reduction of speed in their motor when using the proposed serrated blades.

12.4.3 Automatic Noise Canceling

Another way to mitigate noise is by carrying out a digital signal processing technique known as automatic noise canceling. A version of this technique that is applicable to cancelling drone noise is exemplified in Figure 12.3.

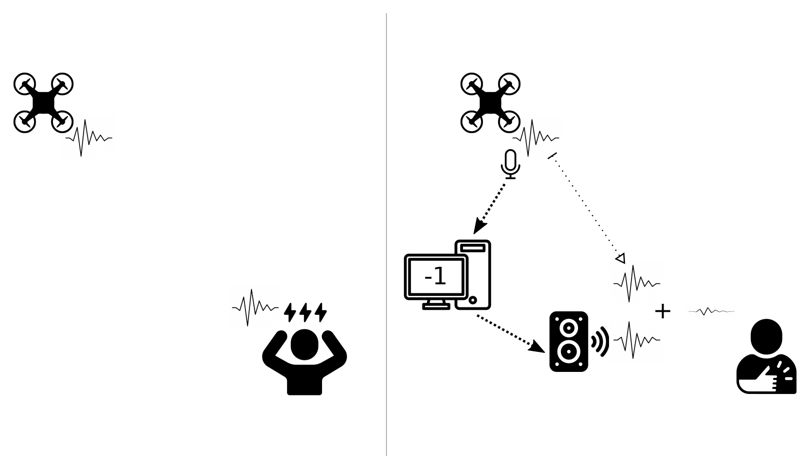

FIGURE 12.3
Example of automatic noise canceling. Left: no noise canceling. Right: noise canceling is carried out.

As it can be seen, to carry out this technique it is necessary to, first, capture the noise that is wished to be canceled, which implies that the drone needs to carry on-board microphones to capture the noise created by its motors. Then, simultaneously, a phased version of the noise (basically, the waveform of the noise multiplied by a factor of -1) is to be reproduced close to the location where it is desired that the noise is mitigated. The result being that, by the additive nature of the acoustic interaction of two signals, a destructive interference occurs that results in the noise being canceled. There is an important limitation to this: the noise source and its phased reproduction need to be perfectly out of phase so that the cancellation occurs. This requires proper synchronization of the captured signal and the reproduced signal, as well as very low latency reproduction.

The work of [5] uses several microphones on board the UAV which capture the noise in different directions; the authors call these captured signals 'secondary sources'. These are modeled using spherical sector harmonics, and by using a matching process, are then canceled out. Simulated results showed between 20 and 35 dB of noise reduction.

The work in [28] actually uses the tonal nature of the UAV noise to cancel it. For each of the tones of the UAV noise, a pitch shifting technique is employed and is controlled by a closed-loop controller that monitors the output signal. The system was able to reduce the noise energy by 25%, up to 44%.

The authors of [3] proposed a noise-free recording process that uses an on-board microphone to capture the UAV noise, and a second microphone that captures the noisy target speech (which is to be filtered). The latter is expected to be mixture of the target speech and a version of the UAV noise. In the first phase of the process, an automatic noise cancellation process is carried out, with a speaker reproducing a phased version of the UAV noise in real-time. In the second phase, which seems to be carried out in an offline manner, subtracts the frequency-domain version of the recorded UAV noise from the frequency-domain version of the output of the first phase, as a type of post-filter. The final result showed a high similarity (67.5%) to the clean version of the target speech with almost a complete cancellation of the UAV noise.

As it can be seen, there are very few works that aim to cancel the UAV noise via digital signal processing techniques, which implies that it is a foundling area of research. However, it is important to note that most, if not all, of these techniques will require additional audio hardware, both on-board the UAV and on the ground (near the population sensitive to the UAV noise), to capture and reproduce audio signals. All of which may be non-viable in some scenarios.

12.5 Discussion and Final Remarks

The growth of the drone market for leisure and civilian applications has raised concerns regarding public acceptance of the drone technology. These concerns include safety, health, and noise pollution issues. Sound pressure level (SPL) is an important indicator used to measure drone acoustics. Exposure to high SPLs can have detrimental effects on human hearing, making it important to monitor and control noise levels. The SPL of a drone depends on several factors and ranges from 60 to over 100 dB. The lack of specific standards for drone noise underscores the need to establish means to measure its effects and develop guidelines for deployment in urban settings.

Public acceptance of drones is affected by factors such as annoyance caused by sound levels, societal safety, and environmental impacts. While public opinions on drones vary across time and countries, surveys indicate a growing trend of acceptance for urban air mobility, with between half and three-quarters of the public supporting this idea. According to [10], societal concerns related to drones include noise impact, impacts on animals and flora, visual pollution, and loss of privacy or intrusion. Similarly, the works in [12, 17] evaluated public acceptance of civilian drones for daily-life applications.

The most pressing concerns were related to criminal activity and privacy breaches, followed by liability, safety, and damages, with noise being the least concern. Catastrophe response, research work, and civil protection had the highest acceptance rates, while parcel delivery and air taxi had low acceptance rates due to safety concerns. Personal use of drones showed a similar trend with higher approval for non-profit and social benefit applications. Noise concerns, age, and gender influenced attitudes towards drones, with advanced air mobility for passenger transport having little acceptance. The study calls for establishing regulations for noise management to ensure sustainable UAM.

The continuous presence of noisy drones flying over homes and workplaces may lead to decreased public acceptance due to safety and health concerns, similar to living or working near airports. The World Health Organization (WHO) recommends avoiding exposure to levels above 45 dB, and more than 40 dB at night. To address concerns over drone noise, research groups are developing evaluation frameworks and metrics to understand the effects of drone noise on humans and animals under various scenarios. Research has also shown that different species have varying levels of sensitivity to noise at different frequencies generated by drones, emphasizing the need to reduce drone impact on target species and inform future drone use guidelines.

In order to understand the basis for noise analysis, in this chapter, we have also dedicated a section to Noise modeling, describing essential concepts such as Noise Pressure Level, Tonal and Broadband noise found in several works touching on the topic of drone noise and measurement for the public acceptance [39, 43, 35, 44, 45].

We have also delved into noise mitigation techniques, seeking to discuss recent works on reducing or canceling the noise emitted by a drone, particularly multi-rotor drones. As mentioned in [10], noise mitigation may reduce noise annoyance, which may not be essential for one or a few drones that add noise to existing soundscapes. However, UAM envisages using a significant number of drones whose noise may cause considerable distress to human and animal life, equally or even more than conventional traffic noise.

While drone noise may cause discomfort in civilian applications, the emitted noise can be used to detect drones [37, 7, 25] when visual or radar systems may not be sufficient [51]. In contrast, developing silent or stealth drones may compromise the safety and accountability of drone operations in civilian spaces.

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